

# **BUSINESS INFORMATION SYSTEMS- PART 2**

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# Objectives

- Explain the need for system analysis and design for business information system.
- Compare the underlying assumptions and uses of a predictive and an adaptive system development life cycle (SDLC).
- Explain what makes up a system development methodology—the SDLC as well as models, tools, and techniques.
- Describe methodologies, models, tools, and techniques of system analysis and development to be used in business information system.
- Describe the key features of Agile development.
- Describe the business systems development life cycle.
- Discuss the techniques of analyzing business data using Artificial Intelligence (AI).



# Need for Systems Analysis and Design

- Security is a critical factor and a challenge when developing new systems
- Installing a system without proper planning leads to great user dissatisfaction and frequently causes the system to fall into disuse
- Systems analysis and design lends structure to the analysis and design of information systems
- User involvement throughout a systems project is critical to the successful development of computerized systems



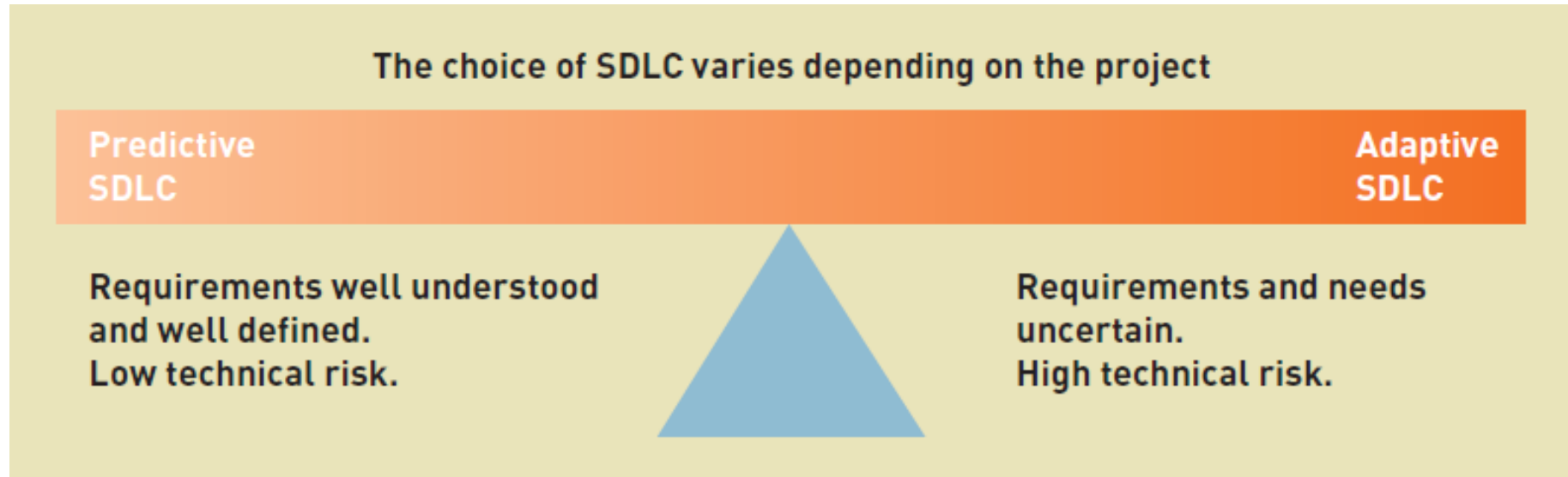
# The System Development Life Cycle (SDLC)

- There are two general approaches to the SDLC
- Predictive Approach to the SDLC
  - Waterfall model
  - Assumes the project can be planned in advance and that the information system can be developed according to the plan
  - Requirements are well understood and/or low technical risk
- Adaptive Approach to the SDLC
  - Iterative model (as see in this text)
  - Assumes the project must be more flexible and adapt to changing needs as the project progresses
  - Requirements and needs are uncertain and/or high technical risk



# The System Development Life Cycle (SDLC)

- Most projects fall on a continuum between Predictive and Adaptive

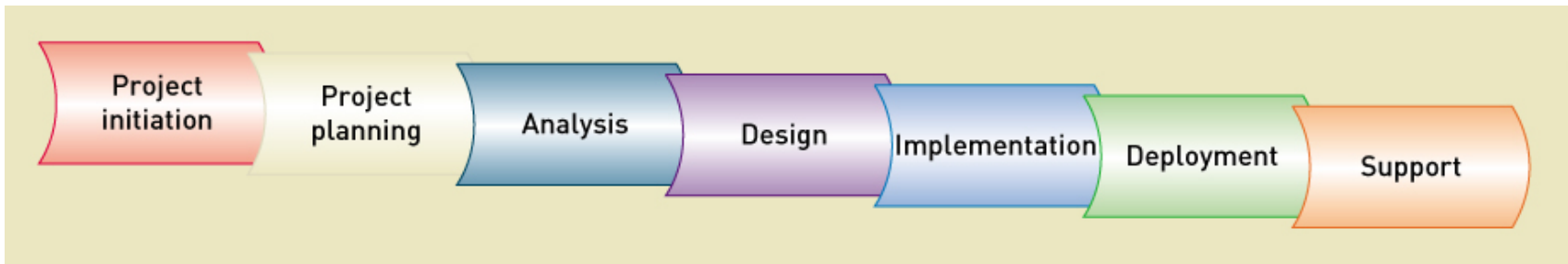


# Traditional Predictive SDLC

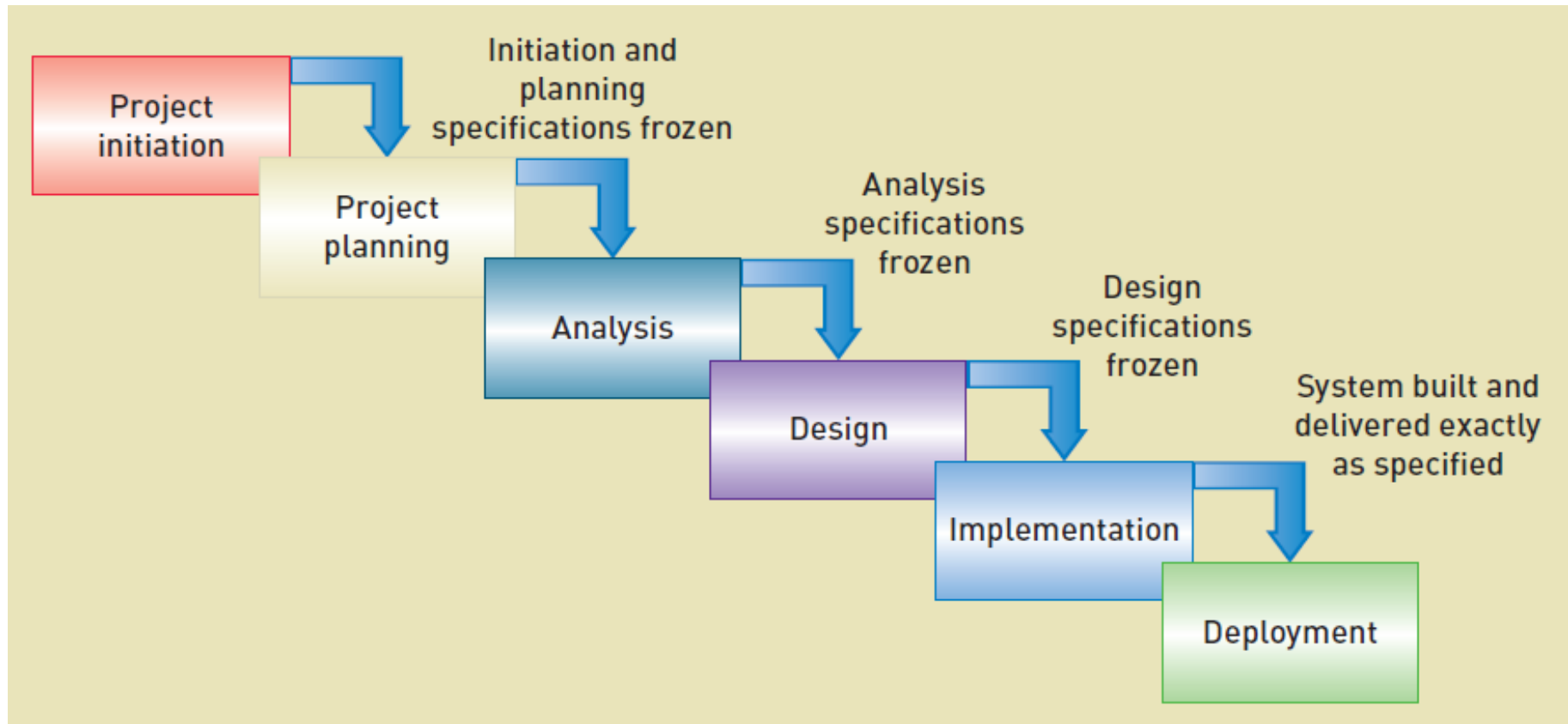
- Earlier approach based on engineering
- Typically have sequential *Phases*
  - Phases are related groups of development activities, such as planning, analysis, design, implementation, and deployment
- Waterfall model
  - SDLC that assumes phases can be completed sequentially with no overlap or iteration
  - Once one phase is completed, you fall over the waterfall to the next phase, no going back



# Traditional Information System (IS) Development Phases



# Waterfall Predictive SDLC





# Newer Adaptive SDLC

- Emerged in response to increasingly complex requirements and uncertain technological environments
- Always includes iterations where some of design and implementation is done from the beginning
- Many developers claim it is the *only* way to develop information systems
- Many IS managers are still sceptical due to apparent lack of overall plan



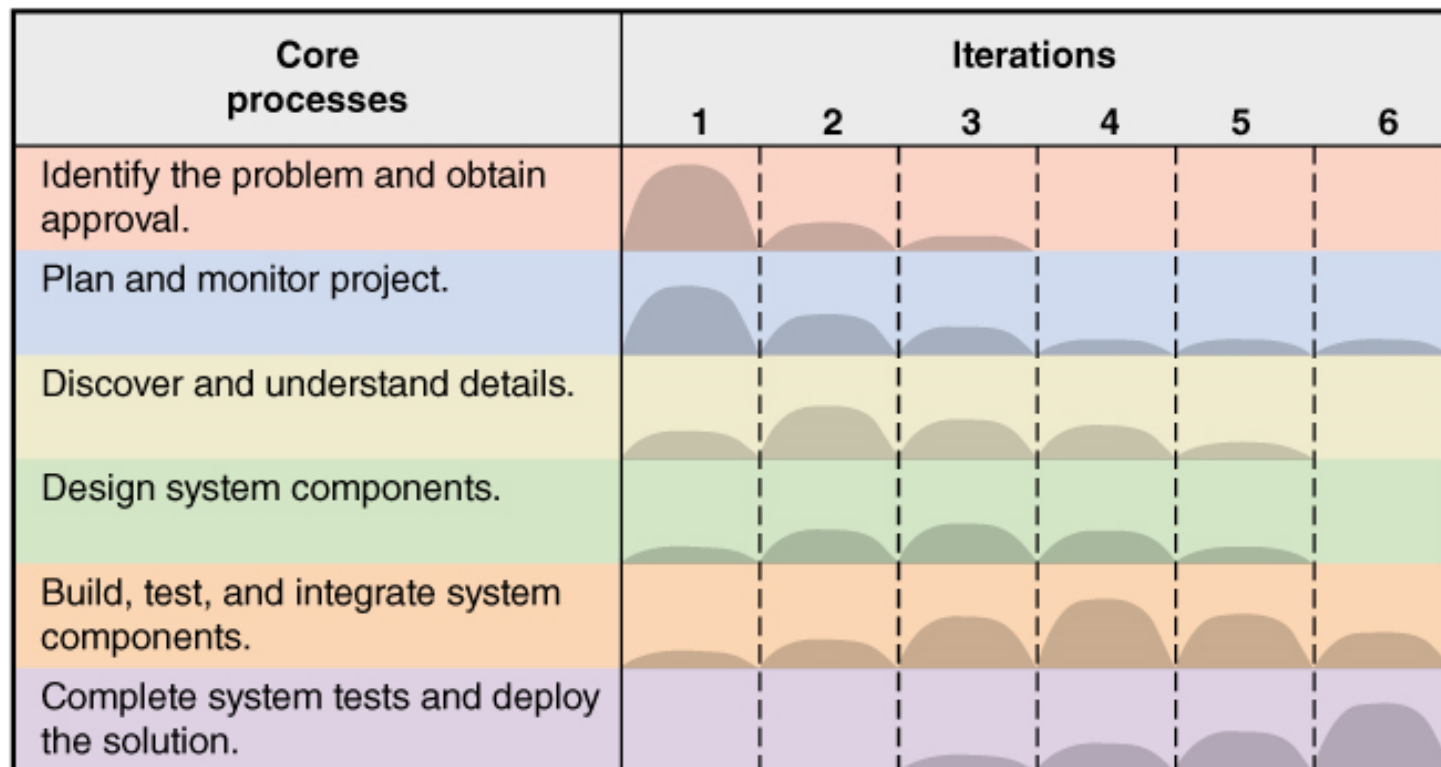
# Adaptive Approaches

- Incremental development
  - Completes portions of the system in small increments and integrated as the project progresses
  - Sometimes considered “growing” a system
- Walking Skeleton
  - The complete system structure is built first, but with bare-bones functionality



# A Generic Adaptive Approach

- Six Core Processes go across iterations
- Multiple Iterations as required



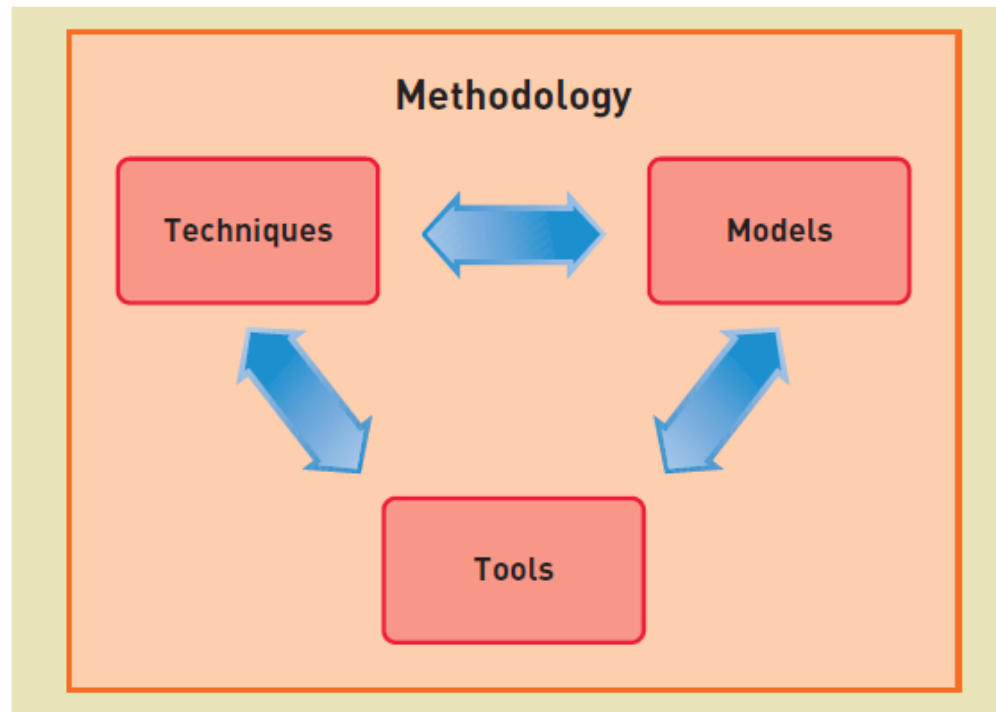
# Methodologies, Models, Tools, and Techniques

- Methodologies
  - Provides guidelines for every facet of system development: What to do when, why and how
  - Specifies an SDLC with activities and tasks
  - Specifies project planning and project management models and reporting
  - Specifies analysis and design models to create
  - Specifies implementation and testing techniques
  - Specifies deployment and support techniques
- Other term used is *System Development Process*



# Methodologies

- A *Methodology* includes a collection of techniques that are used to complete activities and tasks, including modelling, for every aspect of the project



# Methodologies, Models, Tools, and Techniques

- Model
  - An abstraction of an important aspect of the real world.
  - Makes it possible to understand a complex concept by focusing only on a relevant part
  - Each model shows a different aspect of the concept
  - Crucial for communicating project information
- In IS, some models are of system components that will be developed
- Other models are used to manage the development process



# Methodologies, Models, Tools, and Techniques

## Some models of system components

- Use case diagram
- Domain model class diagram
- Design class diagram
- Sequence diagram
- Package diagram
- Screen design template
- Dialog design storyboard
- Entity-relationship diagram (ERD)
- Database schema

## Some models used to manage the development process

- Gantt chart
- Organizational hierarchy chart
- Financial analysis models—NPV, payback period
- System development life-cycle model
- Stakeholders list
- Iteration plan



# Methodologies, Models, Tools, and Techniques

- Tools
  - Software applications that assists developers in creating models or other components required for a project
- Integrated Development Environment (IDE)
  - set of tools that work together to provide a comprehensive development environment
- Visual Modelling Tools
  - Tools to create graphical models





# Methodologies, Models, Tools, and Techniques

- Some typical tools

Project management application

Drawing/graphics application

Word processor/text editor

Visual modeling tool

Integrated development environment (IDE)

Database management application

Reverse-engineering tool

Code generator tool



# Methodologies, Models, Tools, and Techniques

- Technique
  - A collection of guidelines that help an analyst complete an activity or task
  - Learning techniques is the key to having expertise in a field

Strategic planning techniques  
Project management techniques  
User-interviewing techniques  
Data-modeling techniques  
Relational database design techniques  
Structured programming techniques  
Software-testing techniques  
Process modeling techniques  
Domain modeling techniques  
Use case modeling techniques  
Object-oriented programming techniques  
Architectural design techniques  
User-interface design techniques



# Agile Development: 4 Core Philosophy

- Agile Values in “Manifesto for Agile Software Development”
  - Value responding to change
    - quick responses to change and continuous development following a plan
- Value individuals and interactions
  - self-organization and motivation over processes and tools

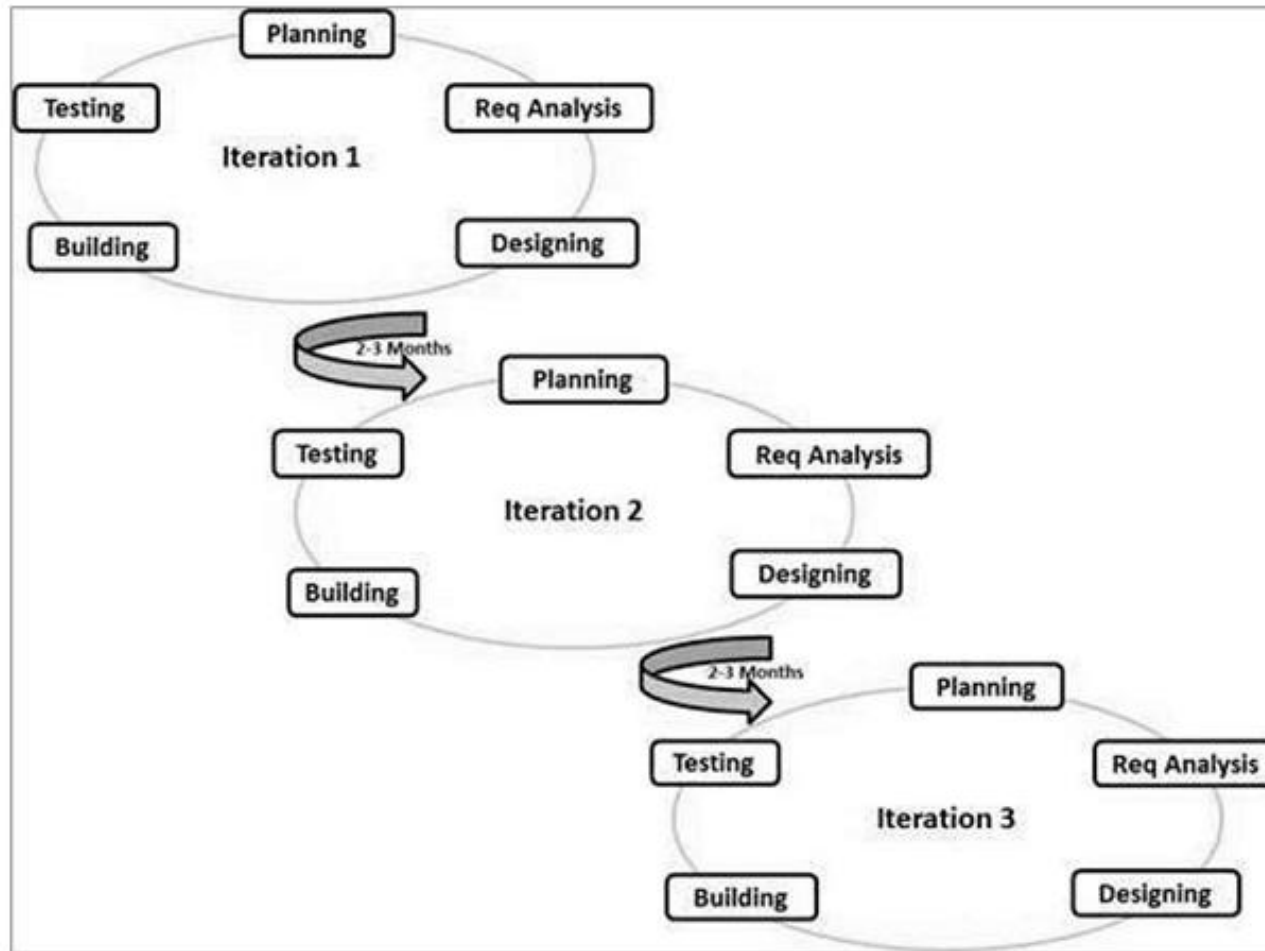


# Agile Development: 4 Core Philosophy

- Agile Values in “Manifesto for Agile Software Development”
  - Value working software
    - demo working software is considered the best means of communication with the customers to understand their requirements, instead of just depending on documentation
  - Value customer collaboration
    - the requirements cannot be gathered completely in the beginning of the project due to various factors, continuous customer interaction is very important to get proper product requirements

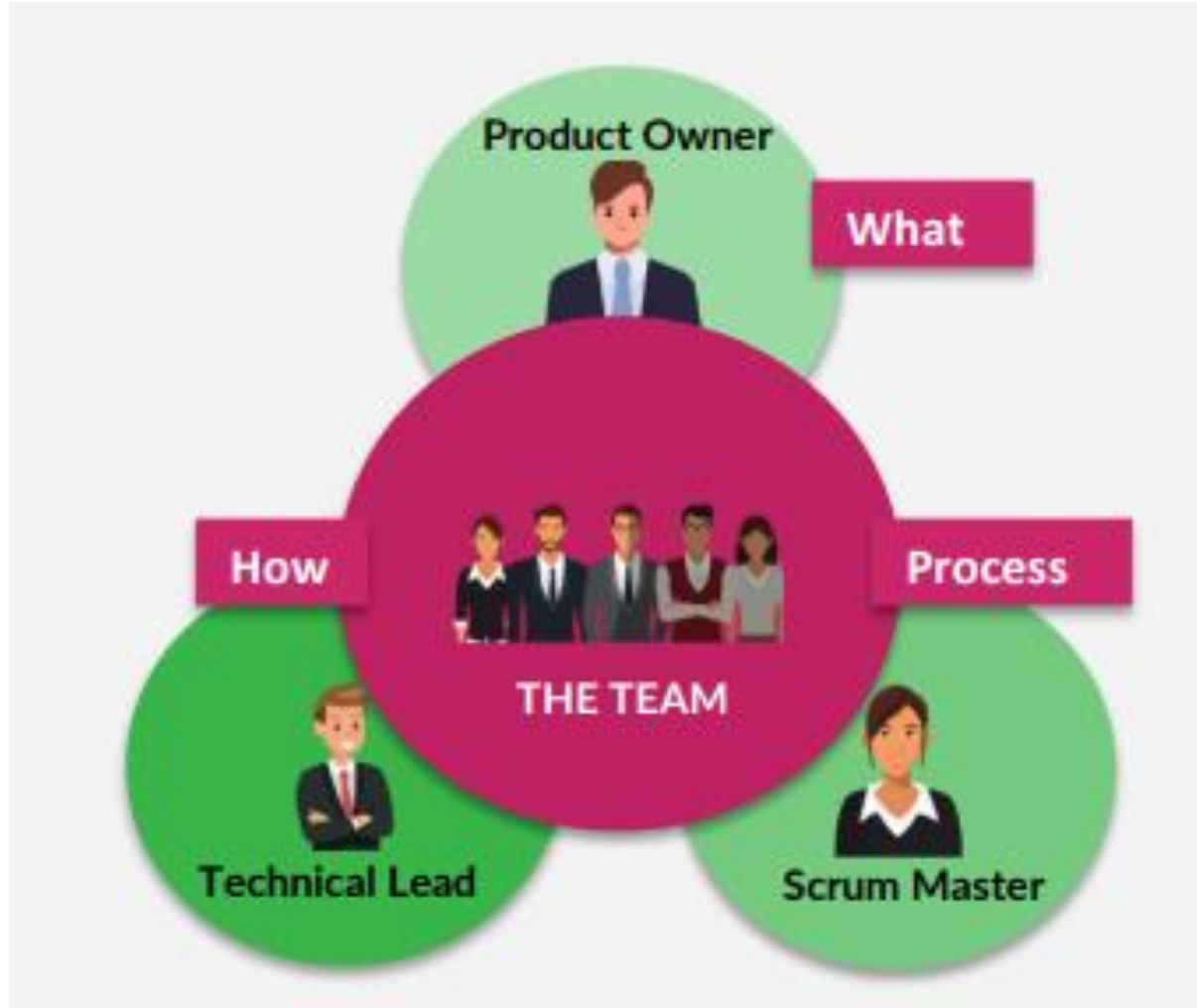


# Agile Development



# Agile Development -Terminology

UNCLASSIFIED - NON CLASSIFIÉ



# Agile Development -Terminology

## Who is The ScrumMaster?

- ❖ Accountable for the Process
- ❖ Builds empowered self organizing teams
- ❖ Is a Servant Leader
- ❖ Removes impediments
- ❖ Creates visibility, measurement, accountability and fun
- ❖ Strong facilitator
- ❖ Encourages collaboration and open communication



# Agile Development

## Who is The Product Owner?

- ❖ One Person , accountable for **Backlog and Scope**
- ❖ **Prioritizes** the backlog
- ❖ Communicates the **Vision** of the product
- ❖ Defines **conditions of satisfaction**
- ❖ **Accepts/rejects** stories
- ❖ **Accountable** for project success
- ❖ Accountable for **ROI**
- ❖ Helps define '**Done**'
- ❖ Solicits **Customer** and





# Agile Development -Terminology

## Who is The Team?

- ❖ Cross-functional team accountable for **delivering business value** iteratively
- ❖ **Engaged, Motivated, Self Organizing, Collaborative**
- ❖ **Commits** to work
- ❖ **Inspects & Adapts**
- ❖ **Plans and re-plans**
- ❖ Delivers '**Done**' stories each iteration
- ❖ Uses TDD, Automation and Best Practices



# Agile Development -Terminology

## Glossary

**BACKLOG:** one list containing all stories

**RELEASE PLAN:** rough schedule of iterations

**RELEASE:** Moving 'Done' stories to production

**BURN UP CHART:** demonstrates visually how many points the team got 'Done'

**VELOCITY:** how many points the team got 'Done' in an iteration.

**STORY:** a description of a small valuable customer requirement

**FEATURE/THEME:** grouping of related stories

**TASKBOARD:** where the team tracks their tasks visibly.

**SCRUM MASTER:** owns the process, leads the team by empowering them, removes impediments, tracks progress.

**TEAM:** cross-functional group working together to get a story 'Done'.

**STORY POINTS:** a relative measure of complexity for a story.



# Agile Development -Terminology

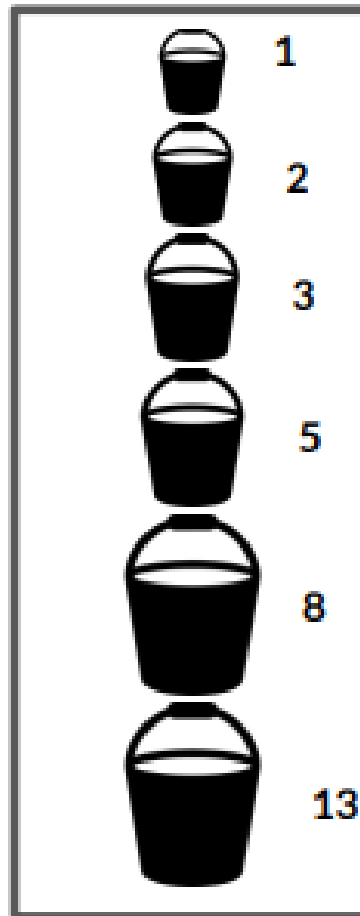
## A Story Should Be:

- ❖ Understandable
- ❖ Independent
- ❖ Negotiable
- ❖ Valuable
- ❖ Estimatable
- ❖ Small
- ❖ Testable

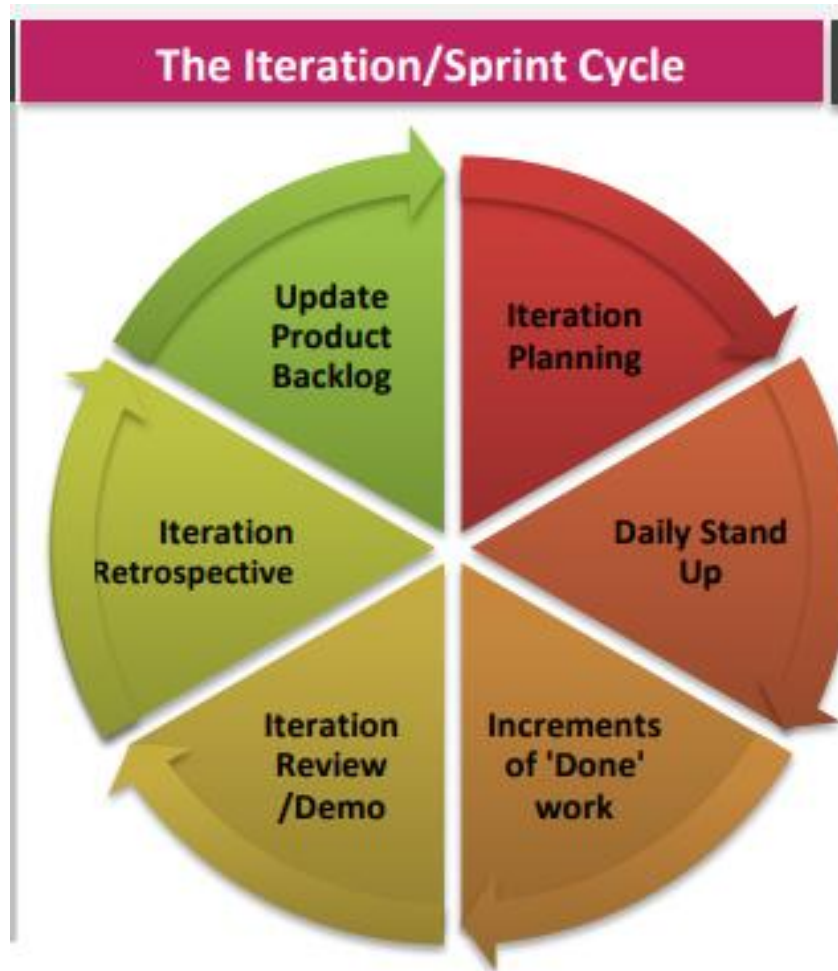


# Agile Development -Terminology

## Story Points



# Agile Development -Terminology



# Agile Development -Terminology

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## 5 SCRUM EVENTS

### THE SPRINT

**PURPOSE** – The sprint is a fixed length event up to a month in length where all the work is completed to turn ideas into value. Considered the heartbeat of Scrum.



1. Every Sprint should be the **SAME LENGTH**.



2. As soon as one Sprint ends, the **NEXT SPRINT BEGINS**.



3. The Sprint is a container for all the other **SCRUM EVENTS**.

### SPRINT PLANNING

**PURPOSE** - Understand WHY the Sprint is valuable, WHAT prioritized items the team will work on and HOW the team will complete the work.



1. The **SCRUM TEAM IS RESPONSIBLE** for planning each backlog item and taking on a realistic amount of work based on their capacity and past performance.



2. The Scrum Team **PLANS THE WORK TOGETHER** with the goal of completing the work together.



3. The Sprint Goal, selected Backlog Items and plan for delivering them is called the **SPRINT BACKLOG**.

### DAILY SCRUM

**PURPOSE** – Inspect progress toward the sprint goal, coordinate efforts and adapt plans.



1. The Daily Scrum is for the **DEVELOPERS** to improve communication and decision-making.



2. The format can vary but the **FOCUS** is on hitting the **SPRINT GOALS**.



3. The meeting should last **LESS THAN 15 MINUTES** and be held at the same time and place every working day of the sprint.

### SPRINT REVIEW

**PURPOSE** - Demonstrate progress, inspect the team results and get feedback for future adaptations.



1. The Scrum Team should show actual working **RESULTS FROM THE USER'S PERSPECTIVE**. Don't show lines of code or PowerPoint.



2. **GET ORGANIZED**, start on time and be succinct. Plan for 4-5 minutes per completed backlog item and leave time for engagement with stakeholders.



3. **EXPECT FEEDBACK** including new requests.

### RETROSPECTIVE

**PURPOSE** - Allow the Scrum team to pause, reflect and plan ways to improve team quality and effectiveness.



1. Retrospectives are **FOR THE SCRUM TEAM** and the team decides who should attend.



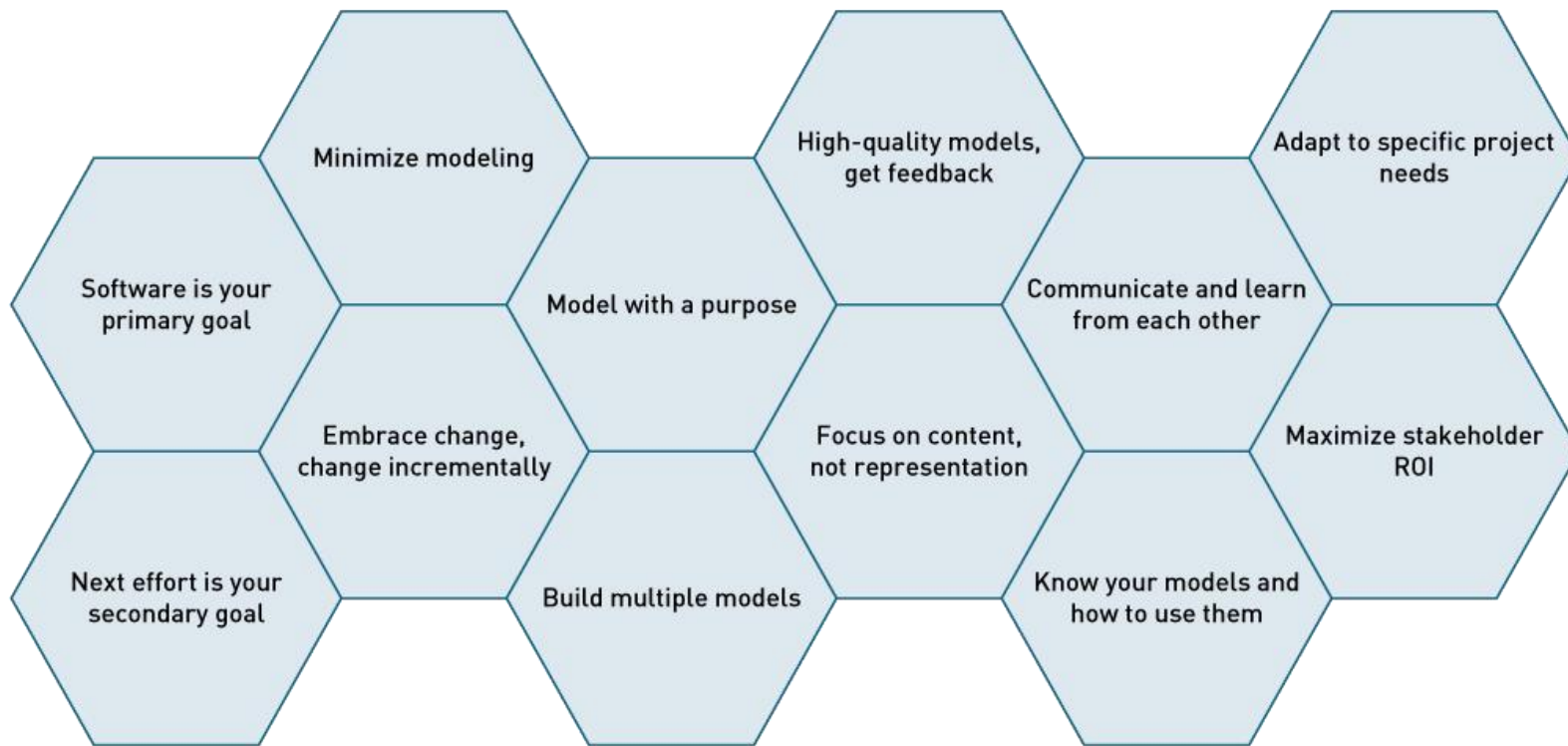
2. **NO BLAMING OR COMPLAINING**. Assume that everyone did the best they could under the circumstances.



3. Use **ROOT CAUSE ANALYSIS** to go beneath the surface. Select just one or two improvement actions each sprint.

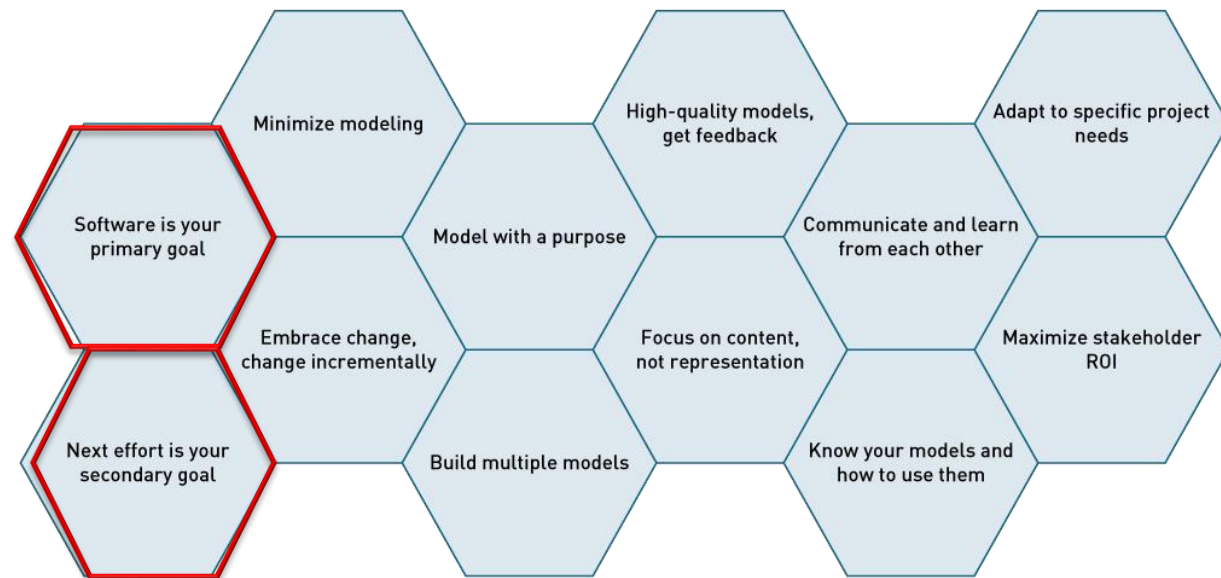
# Agile Modeling Principles

- Agile Modeling (AM) – 12 Principles
  - A philosophy – build only necessary models that are useful and at the right level of detail



# Agile Principles

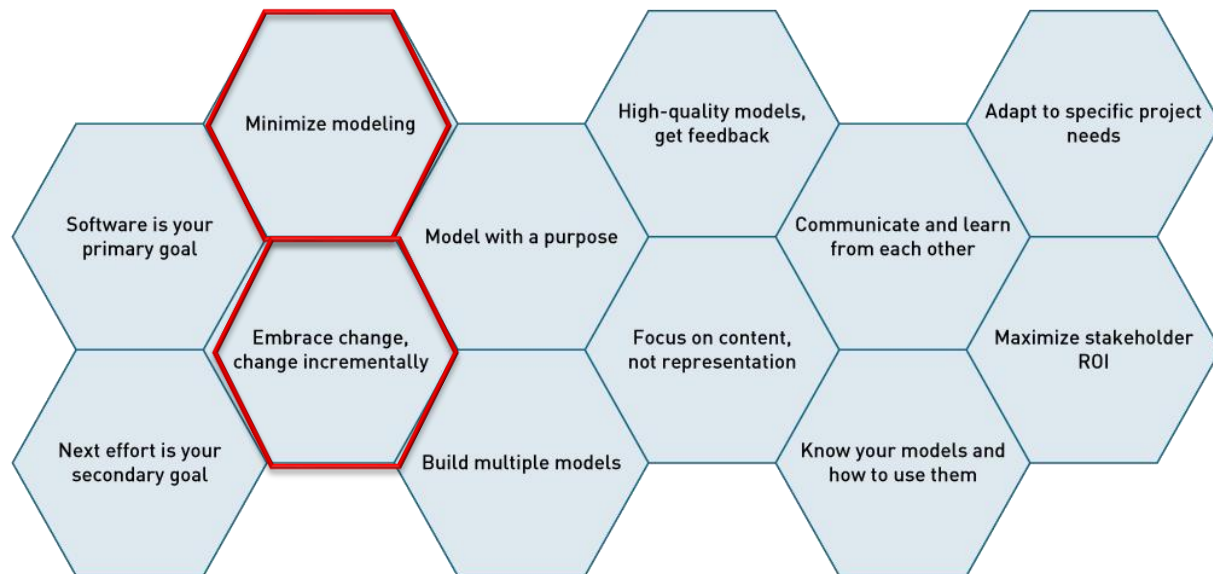
- Develop software as the primary goal
  - Don't get distracted by documentation or models
- Enable the next effort as your secondary goal
  - Be aware of next step versions or revisions





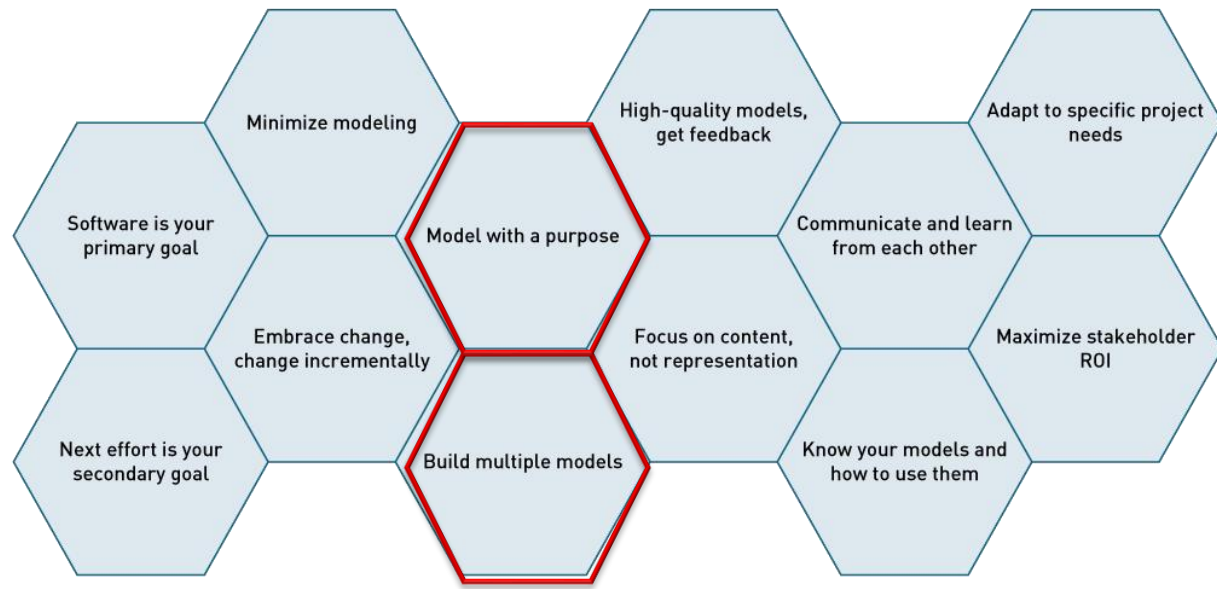
# Agile Principles

- Minimize your modeling activity
  - Only build what helps move the project forward
- Embrace change and change incrementally
  - Take small steps that keep you on-track and that can be reversed if necessary



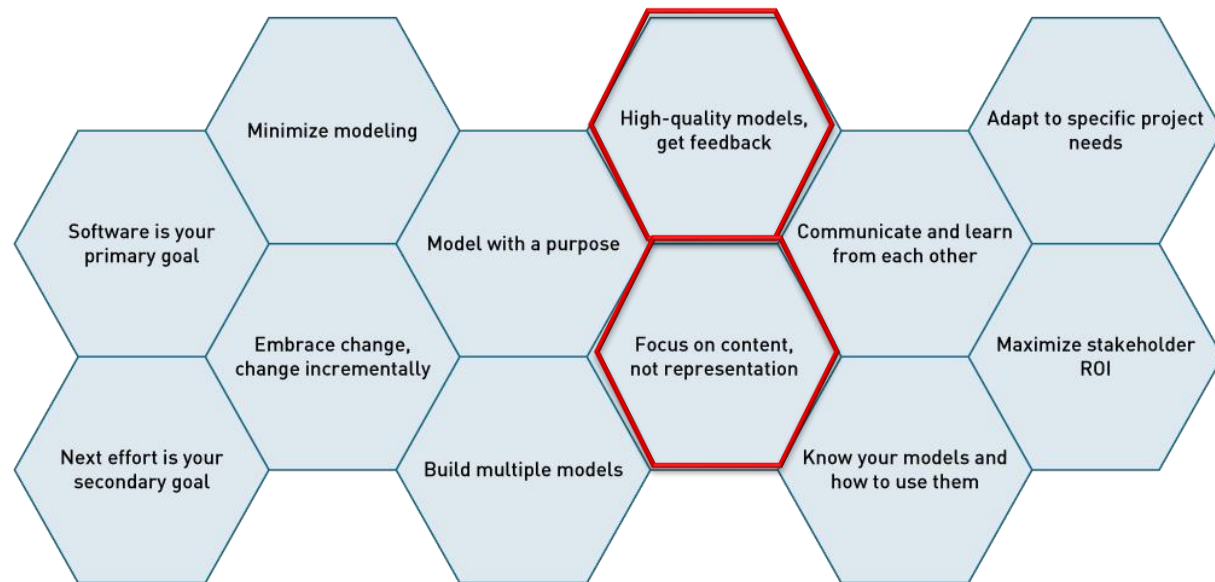
# Agile Principles

- Model with a purpose
  - Model to understand
  - Model to communicate
- Build multiple models
  - Look at problems from different perspectives



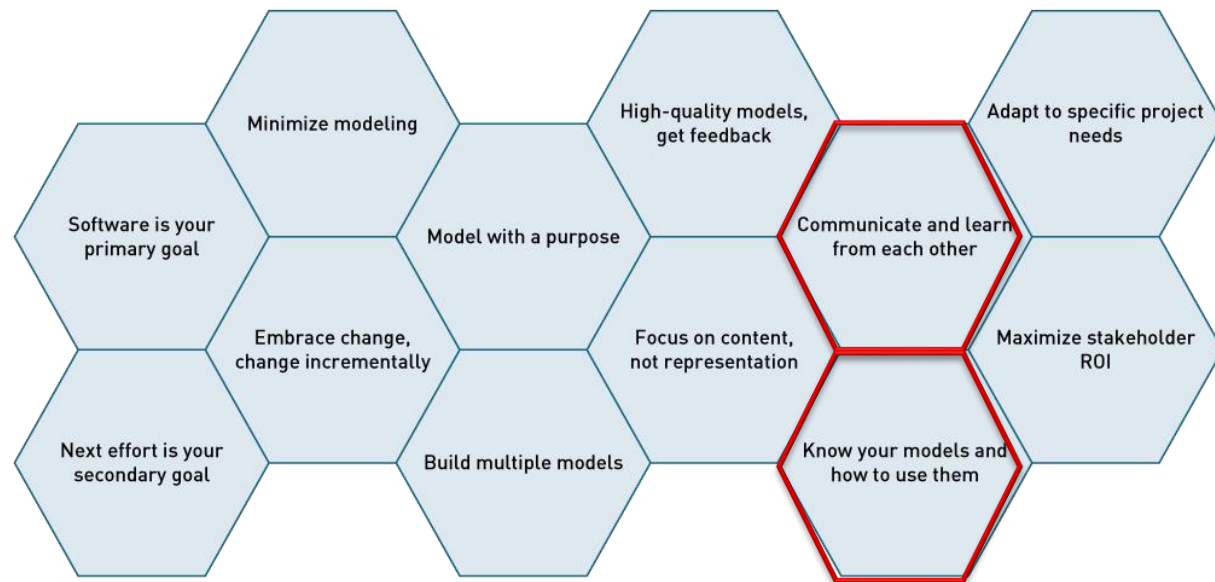
# Agile Principles

- Build high-quality models and get feedback
- Focus on content rather than representation
  - Informal hand-drawn models are sometimes okay
  - Always focus on stakeholder needs



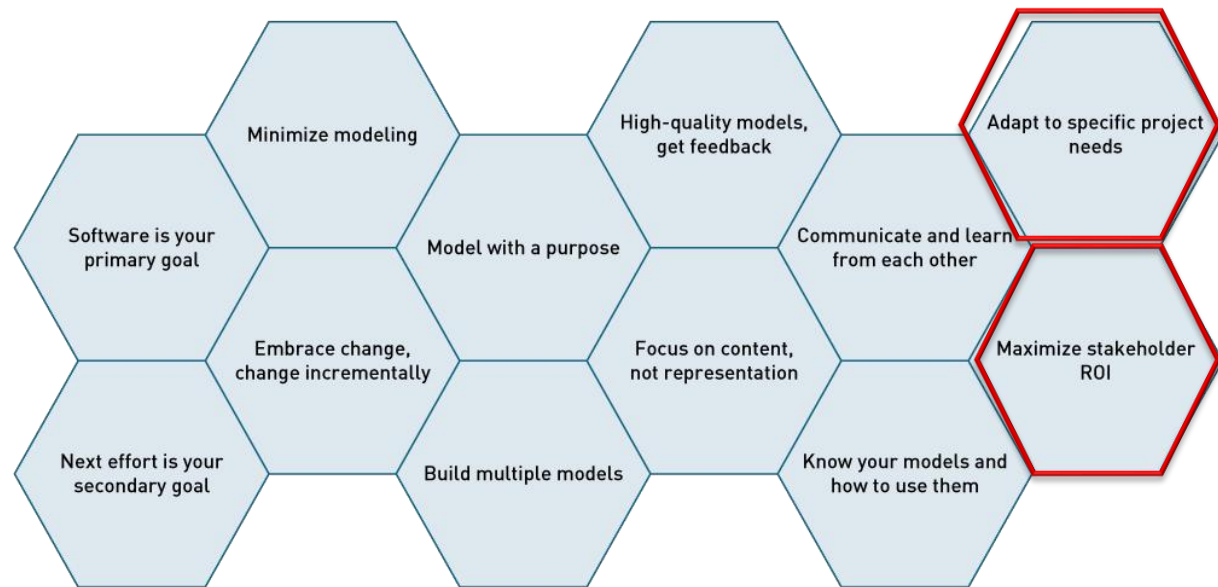
# Agile Principles

- Learn from each other with open communication
- Know your models and how to use them



# Agile Principles

- Learn from each other with open communication
- Know your models and how to use them



# Building an E-commerce Business Site: In the context of Systems Analysis

Most important management challenges:

1. Developing a clear understanding of business objectives
2. Knowing how to choose the right technology to achieve those objectives
  - Main factors to consider
    - Management
    - Hardware architecture
    - Software
    - Design
    - Telecommunications
    - Human resources



# The Business Systems Development Life Cycle

- Methodology for understanding business objectives of a system and designing an appropriate solution
- Five major steps
  - Systems analysis/planning
  - Systems design
  - Building the system
  - Testing
  - Implementation



# System Analysis/Planning

- Business objectives
  - List of capabilities you want your site to have
- System functionalities
  - List of information system capabilities needed to achieve business objectives
- Information requirements
  - Information elements that system must produce in order to achieve business objectives





# System Analysis, Business Objectives, System Functionalities, and Information Requirements for a Typical E-commerce Site (1 of 2)

| Business Objective                | System Functionality         | Information Requirements                                                                                              |
|-----------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Display goods                     | Digital catalog              | Dynamic text and graphics catalog                                                                                     |
| Provide product information       | Product database             | Product description, stocking numbers, inventory levels                                                               |
| Personalize/customize product     | Customer on-site tracking    | Site log for every customer visit; data mining capability to identify common customer paths and appropriate responses |
| Engage customers in conversations | On-site blog; user forums    | Customer comments captured by software with blogging and community forum functionality                                |
| Execute a transaction             | Shopping cart/payment system | Secure credit card clearing; multiple payment options                                                                 |
| Accumulate customer information   | Customer database            | Name, address, phone, and e-mail for all customers; online customer registration                                      |



# System Analysis, Business Objectives, System Functionalities, and Information Requirements for a Typical E-commerce Site

## (2 of 2)

| Business Objective                    | System Functionality                                                  | Information Requirements                                                                       |
|---------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Provide after-sale customer support   | Sales database                                                        | Customer ID, product, date, payment, shipment date                                             |
| Coordinate marketing/advertising      | Ad server, e-mail server, e-mail, campaign manager, ad banner manager | Site behavior log of prospects and customers linked to e-mail and banner ad campaigns          |
| Understand marketing effectiveness    | Site tracking and reporting system                                    | Number of unique visitors, pages visited, products purchased, identified by marketing campaign |
| Provide production and supplier links | Inventory management system                                           | Product and inventory levels, supplier ID and contact, order quantity data by product          |



# Systems Analysis: Hardware and Software Platforms

- System design specification
  - Description of main components of a system and their relationship to one another
- Two components of system design:
  - Logical design
    - Data flow diagrams, processing functions, databases
  - Physical design
    - Specifies actual physical, software components, models, and so on



# Building the System: In-House Versus Outsourcing

- Outsourcing: Hiring vendors to provide services involved in building site
- Build own versus outsourcing
  - Build your own requires team with diverse skill set; choice of software tools; both risks and possible benefits
- Host own versus outsourcing
  - Hosting: Hosting company responsible for ensuring site is accessible 24/7, for monthly fee
  - Co-location: Firm purchases or leases web server (with control over its operation), but server is located at vendor's facility



# Testing the System

- Testing
  - Unit testing
  - System testing
  - Acceptance testing
  - A/B testing (split testing)
  - Multivariate testing



# Implementation, Maintenance, and Optimization

- Systems break down unpredictably
- Maintenance is ongoing
- Maintenance costs: Similar to development costs
  - A \$40,000 e-commerce site may require \$40,000 annually to upkeep
- Benchmarking



# Web Development Methodologies for Business Information Systems

- Prototyping
- Agile development
- DevOps
- Component-based development
- Web services
  - Service-oriented architecture (SOA)
  - Microservices



# Simple versus Multi-Tiered Website Architecture

- System architecture
  - Arrangement of software, machinery, and tasks in an information system needed to achieve a specific functionality
- Two-tier
  - Web server and database server
- Multi-tier
  - Web application servers
  - Backend, legacy databases





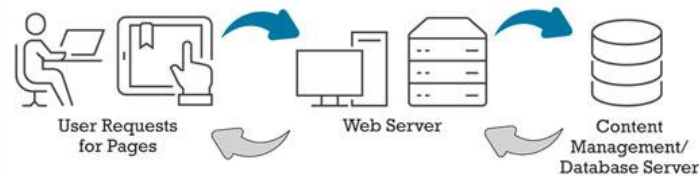
# Two-Tier and Multi-tier E-commerce Site Architectures

FIGURE 4.10

## TWO-TIER AND MULTI-TIER E-COMMERCE SITE ARCHITECTURES

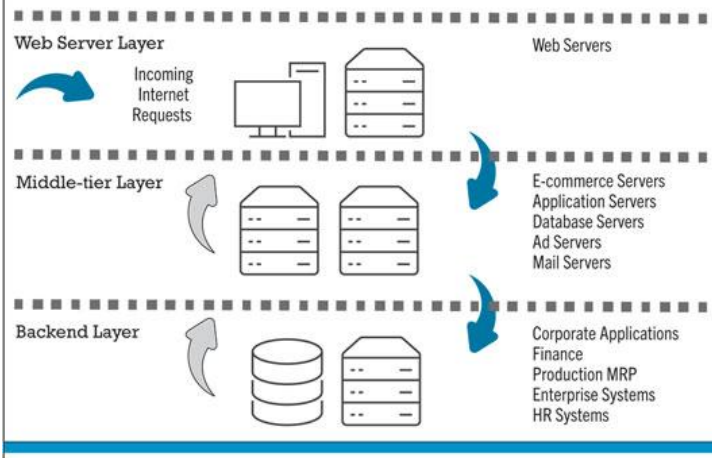
### A. Two-tier Architecture

*In a two-tier architecture, a web server responds to requests for web pages and a database server provides backend data storage.*



### B. Multi-tier Architecture

*A physical design describes the hardware and software needed to realize the logical design.*



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# Choosing Hardware

- Hardware platform
  - Underlying computing equipment needed for e-commerce functionality
- Objective
  - Enough platform capacity to meet peak demand without wasting money
- Important to understand the factors that affect speed, capacity, and scalability of a site



# The Eight Most Important Factors in Successful E-commerce Business Site Design

| Factor                      | Description                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------|
| Functionality               | Pages that work, load quickly, and point the customer toward your product offerings |
| Informational               | Links that customers can easily find to discover more about you and your products   |
| Ease of use                 | Simple foolproof navigation                                                         |
| Redundant navigation        | Alternative navigation to the same content                                          |
| Ease of purchase            | One or two clicks to purchase                                                       |
| Multi-browser functionality | Site works with the most popular browsers                                           |
| Simple graphics             | Avoids distracting, obnoxious graphics and sounds that the user cannot control      |
| Legible text                | Avoids backgrounds that distort text or make it illegible                           |



# Developing a Mobile Website and Building Mobile Applications for Business Information Systems

- Types of m-commerce platforms
  - Mobile website
  - Native app
  - Mobile web app
  - Hybrid app
    - Runs inside native container
    - App distribution
    - Based on HTML5, CSS, JavaScript



# Planning and Building a Mobile Presence

- Identify business objectives, system functionality, and information requirements
- Choices
  - Mobile website or mobile web app
    - Less expensive
  - Native app
    - Can use device hardware, available offline



# Unique Features That Must Be Taken Into Account When Designing a Mobile Presence

| Feature      | Implications For Mobile Platform                                                                                                                                                                         |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware     | Mobile hardware is smaller, and there are more resource constraints in data storage and processing power.                                                                                                |
| Connectivity | The mobile platform is constrained by slower connection speeds than desktop websites.                                                                                                                    |
| Displays     | Mobile displays are much smaller and require simplification. Some screens are not good in sunlight.                                                                                                      |
| Interface    | Touch-screen technology introduces new interaction routines different from the traditional mouse and keyboard.<br>The mobile platform is not a good data entry tool but can be a good navigational tool. |



# Artificial Intelligence (AI) in Analyzing Business Data

- **Artificial Intelligence**
  - Computer systems and algorithms that are capable of performing tasks that typically require human intelligence



# Five Key Concepts in Artificial Intelligence (1 of 2)

1. **Machine learning (ML)**— a subset of artificial intelligence (AI) that focuses on algorithms and statistical models that enable computers to learn from data and make predictions or decisions without explicit programming
2. **Neural networks**—interconnected nodes that process and transmit information
3. **Natural language processing (NLP)** —trained on inputs that come from human speech and written language.





# Five Key Concepts in Artificial Intelligence (2 of 2)

4. **Computer vision**—enabling machines to understand and interpret visual information from images or videos
5. **Robotics**—machines that can interact with the physical world



# References:

- [https://www.tutorialspoint.com/sdlc/sdlc\\_agile\\_model.htm](https://www.tutorialspoint.com/sdlc/sdlc_agile_model.htm)
- <https://agilityinsights.ai/wp-content/uploads/2017/05/Copy-of-Handout-Agile-Scrum-Cheat-Sheet.pdf>

